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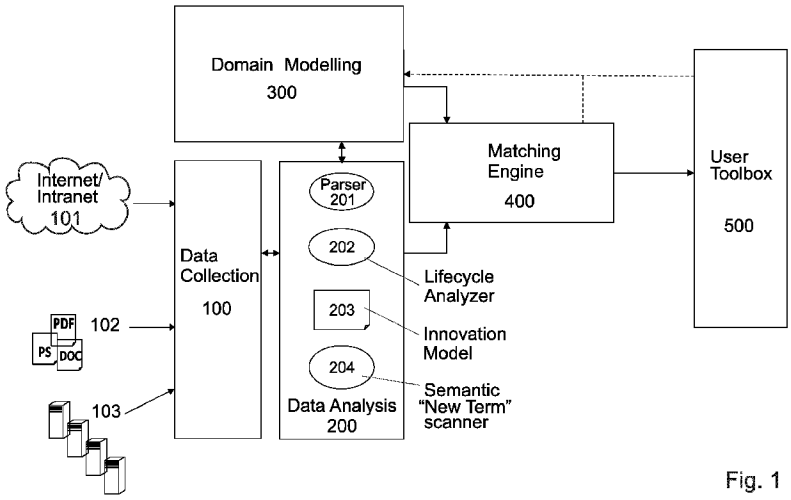

Fig. 1

(57) Abstract: The present invention relates to a method of computer assisted innovation. The invention provides a method which can automatically generate suggested innovation opportunities which may then be viewed or otherwise communicated to and analysed by a user. The invention provides a method and apparatus for determining innovation opportunities by selecting one or more terms; determining trend data relating to a selected element; determining an innovation likelihood measure for said selected element in dependence upon said trend data; identifying an innovation opportunity in dependence upon said innovation likelihood measure.

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A method and apparatus for computer assisted innovation

BACKGROUND

a. Field of the Invention

The present invention relates to a method of computer assisted innovation. The invention provides amethodwhich can automatically generatesuggested innovation opportunities which may then be viewed or otherwise communicated to and analysed by a user.

b. Related Art

The way in which commercially valuable ideas are generated within organizations has remained largely untouched by the general trend towards improved efficiency through automation. Innovative thinking has traditionally been a very manual process, sometimes stimulated through the application of psychological techniques, such as brainstorming. These techniques bring limited improvement, as they are limited to the pre-existing knowledge of the individuals involved and frequently lack any formal structure.

Another approach to generate innovative ideas is the use of a problem analysis tools and methodologies, some of which are enabled by computer-based technologies that can be applied by a researcher or designer.

Problem analysis tools and methodologies can help a user identify a complex system and identify discrete problems which should be addressed. Here computer systems assist in the application of methods of problem analysis. These include root cause analysis, TRIZ (a Russian acronym for

"TeoriyaResheniyaIzobretatelskikhZadatch"), value engineering, system functional analysis, and system benchmarking. TRIZ is a methodology, tool set, knowledge base, and model-based technology for generating innovative ideas and solutions for problem solving. An example of such a tool, called TechOptimizer(TM), is a computer system marketed by Invention Machine Corporation of Boston, Mass. The technology used in TechOptimizer(TM) to assist in problem analysis is partially described in US 6,056,428 and US 6,202,043.

A key deficiency with problem analysis tools is that while they aid in the identification of specific issues to be addressed, the user of such tools is required to possess a good knowledge about the problem being considered. In US20070112746, "System and Method for Problem Analysis,", there is disclosed a method for providing user specific relevant information to assist in the modelling of problems, wherein the principle aspect of analysis is cause-effect relationships, and wherein a cause or effect statement is automatically reformulated as a natural language query. The query is submitted to a database, and the results of the query are returned, thereby greatly facilitating the process of identifying related cause-effect data.

An alternative approach is described by US5390282, “A process for problem solving using spontaneously self-replicating and self-improving entities”. This invention relates to evolutionary processes useful for problem solving by generating self-replicating and self-improving computer programs.

An approach which is specific to controllers is described by US2004030414, “Method and apparatus for automatic synthesis of controllers”. This invention relates to the field of automatic synthesis of complex structures; more particularly, the automatic synthesis of the topology and parameter values for

controllers and control systems.

Yet a further approach to automated innovation has been investigated by Hewlett-Packard and others such as Pfizer. This is exemplified by the paper "Automated re-invention of six patented optical lens systems using genetic programming", Proceedings 2005 Conference on Genetic and Evolutionary Computation on pages 1953 – 1960 which describes how genetic programming was used in an invention machine to automatically synthesize complete designs for six optical lens systems that duplicated the functionality of previously patented lens systems. The automatic synthesis was done "from scratch"-that is, without starting from a pre-existing good design and without pre-specifying the number of lenses, the physical layout of the lenses, the numerical parameters of the lenses, or the non-numerical parameters of the lenses. One of the six genetically evolved lens systems infringed a previously issued patent; three contained many of the essential features of the patents, without infringing; and the others were non-infringing novel designs that duplicated (or improved upon) the performance specifications contained in the patents.

US6360191 describes an automated design process and apparatus for use in designing complex structures, such as circuits, to satisfy prespecified design goals, using genetic operations.

Natural Selection Inc., San Diego, has also used similar technologies based on Darwinian principles of natural evolution and filed patent application US2006036560, "Intelligently interactive profiling system and method".

US7685118 describes a semantic method using an ontology to solve inventor problems. Different possible solutions are ranked based on their generality.

US6167370 describes a piece of software for semantically analysing documents and for creating new concepts and new ideas.

WO2005060684 suggests a semantic approach to Triz. The user needs to express a problem in terms of a natural language query that contains a contradiction and which is submitted to a semantically indexed database.

US5581663 describes another computer-implemented method for solving problems based on the Triz approach. The method starts from a model of the real-world to identify problems and suggest solutions.

US20080195604 describes an automated method for drafting an invention disclosure based on an initial invention concept. The invention involves semantically parsing an initial invention statement, performing an automated prior art search in patent databases to determine patentability issues, and generate an invention disclosure for filing.

US20010034629 describes a method for facilitating the conception of inventions; mainly based on cooperative work among participants, using guidelines etc.

WO07081519 describes another method for generating ideas and selling inventions.

US7533035 describes a method for evaluating innovations based on several factors, and discarding innovations which don't reach a predetermined threshold.

US20040078192 describes a method for automatically performing a semantic analysis of patents and patent claims, and analysing potential infringement.

However, it will be seen that current tools suffer from one or more of the following problems:they need a problem to be defined, require extensive user interaction and training, tend to produce small incremental improvements of low or zero commercial value,are not reactive to the introduction of new technologies and do not work in real-time. As a result users of these systems have to contend with a number of further problems which may include: staff must be trained to use certain methodologies and tools, patenting opportunities may be lost to competitors who identify the innovation opportunity more quicklyor the potential invention generated by use of the tools is not novel orhas of littleor no value.

Moreover, many prior art systems deliver large number of possible solutions to a problem, with no easy way for the user to determine which of those possible solutions is the best one and in particular which solution will give the best patent opportunities.

Moreover, many prior art systems request a lot of human intervention and are not fully automated. When the prior art systems use computers or other processing means, they usually don't use them in an effective way, and request fast processors and/or lot of memory.

An aim of the present invention is thus to develop a new, computer based approach to generate innovation, using computers and semantic methods for detecting as early as possible innovation opportunities which could not be detected without computer.

By taking a new approach to the generation of innovation opportunities, the proposed invention overcomes a number of the problems of the prior art.

In particular, the present invention provides a more automated method of determining innovation opportunities, in which human intervention is reduced.

The present invention also provides a computer-enabled method of determining innovation opportunities, in which a more efficient use of computing resources is made.

The present invention also provides a computer-enabled method of determining innovation opportunities, with which invention opportunities can be identified which might be difficult to identify with a non-computer-enabled method, other than by chance.

The present invention also aims to use computer and software modules for automatically and efficiently detecting innovation opportunities, based on an automatically, continuously performed analysis of trend data for a plurality of terms, and detection of unexpected changes in the frequency of use of some terms or new terms.

SUMMARY OF THE INVENTION

According to the invention there is provided acomputer-enabled method of determining innovation opportunities comprising the steps of:

a) causing a data collection module in a processing unit to store expressions retrieved from external data sources in a data collection;

b) causing a data analysis module in a processing unit to identify

new expressions among said expressions, based on trend data retrieved for the selected expressions, said trend data comprising use of said expressions in said data source during a period;

c) causing a module in the processing unit to identify an innovation opportunity associated with the identified new expressions; and

d) presentingsaid innovation opportunity for information and analysis by a user.

The trend data indicates the frequency of use of an expression during a period of time, for example the frequency of use of the expression in source documents, and/or the number of searches for this expression in search engines. An innovation likelihood can be computed form the trend data. Aninnovation likelihood measure may be a numerical value indicating the probability of an innovation occurring, or may be some other indicator that an innovation may occur. The computation of innovation likelihood from the trend data may be domain specific to reflect the various pace of development in different technologies.

In one aspect of the invention the new expressions are automatically identified, using a computer-enabled retrieval of expressions from various sources and comparison with a set of previously found expressions. New expressions might be for example new words, or new combinations of words, such as new n-grams, for example new bi-grams, which appeared in the data collection during an observation period. New expressions are more likely associated with a high innovation likelihood, based on their associated trend data. For example, an expression which becomes suddenly more popular in the data collection might be associated, during this period, with a high innovation likelihood. An innovation opportunity might then be identified if this new expression is related to the area of interest; a comparison or match

with terms used in an ontology defining the area of interest might be used for determining if a new expression is related to the area of interest.

The step of identifying the innovation opportunity thusoptionally comprises identifying a match between a new expressionand an element in the stored reference data which describes the technical area of interest.

The stored reference data which describes the technical area of interest may be defined with an ontology. It may comprise a reference framework comprising a plurality of interconnected nodes, each node describing an element, such as a term or list of terms.

According to one aspect, the stored reference data is updated in dependence upon feedback from a user relating to said innovation opportunity.

According to one aspect of the invention, the method comprises the steps of: determining trend data relating to a plurality of expressions; determining a respective innovation likelihood measure for each of said plurality of expressions in dependence upon said trend data; and identifying said innovation opportunity in dependence upon a plurality of said innovation likelihood measures.

Preferably an innovation model is used to generate the innovation likelihood measure.

According to another aspect of the invention there is provided an apparatus for determining innovation opportunities comprising: means for identifying one or more new expressions which appear in a data collection; a lifecycle analyser arranged in operation to determine trend data relating to a selected

new expression; determine an innovation likelihood measure for said selected new expression in dependence upon said trend data; and identify an innovation opportunity in dependence upon said innovation likelihood measure; and presentation means for presentingsaid innovation opportunity for information and analysis by a user.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described, by way of example only, with reference to the drawings, in which:

Figure 1 is an illustration of an apparatus showing componentsor functional modules according to one aspect of the present invention.

Figure 2 is a flow chart illustrating the method of the present invention;

Figure 3 shows an example reference framework (or ontology);

Figure 4 shows popularity data for some semantic expression shown in Figure 3.

Figure 4a shows a graph of innovation likelihood over time for a term, in this example “LED backlit LCD”, in the context of a specific area of interest.

Figure 5 shows an example innovation model;

Figure 6 is a flow chart showing a portion of the method in more detail in one particular embodiment of the invention; and

Figure 7 is an illustration of showing various aspects of the present invention

Figure 8 is an example of screenshot of the lifecycle analyser

DETAILED DESCRIPTION

Figure 1schematically illustrates some components or functional modules of an apparatus according to one aspect of the present invention. The apparatus comprises a data collection component 100 having a database and data collection module (such as a crawler etc), a data analysis component 200, a domain modelling component 300, a matching engine 400 and a user tool box 500.

The apparatus may be built as a processing unit having software thereon programmed to carry out the method steps described in this specification and bring the requested functionalities. The processing unit may be a computer, a server, or other suitable processing equipment, and computer data carriers, such as an optical, magnetic or semiconductor disk or storage system, for storing software modules or components causing the processing unit to carry out the desired steps of the invention. The apparatus is further connected to external data sources, such as the Internet/Intranet 101, file servers 102, and/or database servers 103, for accessing various data sources and storing the resulting data in a database, which may then be interrogated bya data analysis engine 200 in said processing unit.

Web crawlers may be used to collate “unstructured” data from data sources (e.g. blogs, news sources, real-time social networks, HTML/XML published journals/documents) and Web Searches may be used to collect “structured” data – semantically tagged (optionally, with statistical metadata: time, volume). Crawling frequency may be defined to provide frequent updates,

and allow subsequent system function to be provided in real time.

Some data sources 103 provide dedicated feeds which send data out in real-time using a well-defined structure and format (e.g.RSS). Search engine volume data can also optionally be used. Data may also be manually entered and edited by users.

Data sources 101, 102, 103 may be selected either according to their general availability, to the area of interest or with regard to the desired nature of the innovation opportunities to be generated. For example, academic publications may result in innovation opportunities which have a longer time to market, but which also have a greater probability of broader patent claims when compared to industry publications reporting on near term technologies, such as industry journals.

The data retrieved by the data collection module 100 are stored and managed in a database as part of the module100 and split in expressions, such as words, sentences and/or expressions, for example n-grams (combination of n terms) or bi-grams (combinations of two terms). In this specification and in the claims, the word “expression” (or sometimes “term”) is used as a generic for words (such as “LED”), expressions (such as “time-of-flight, etc.), bi-grams (such as “LED backlight”); numbers (such as “12’000”), proper names (such as “Nyquist”), URLs or any other string.

Content filter can be used to recognise and store url, emails, domain names, person names, non ascii characters and punctuations, upper cases etc.; this special content is ignored, converted, or specially marked. New expressionsnot previously included in the database are added along with the number of use of the expression, and date of use. The number of use of already existing

expressionsis updated in the database, and the date of the new use is stored. After each iteration, the database in the data collection component 100 of the apparatusis updated with a new entry such as:

ExpressionT_i, time t_j, number of occurrences of T_i at t_j,

Fig. 4 is a graph which illustrates as example the daily use of two expressions("time of flight" and "led backlight") over a period of time. Those trend data for each expressioncan be directly produced by the lifecycle analyser 202 from the data stored in the database of the data collection component 100.

The apparatus thus stores or uses an index of use of expressionsin data sources over a period of time (for example several months). As will be seen, this database will be used to determine an innovation likelihood related with a specific expression, at specific instants when the number of use of an expression suddenly changes.

The expressions retrieved by the data collection component 100 can be ambiguous. For example,a query "time-of-flight" on a search engine can return irrelevant hits related to flight duration together with relevant link related to "electronic devices" which measure the time of flight of a photon (the intended result). Accordingly the data collection component 100 may classify the content of each data source (for example each URL)within classification systems – such as the International Patent Classification, or any suitable patent or document classification scheme - in order to ensure the information is contextually correct. This classification might be automatic or semi-automatic, based on an automatic, semantically driven classifying module.The classification can be used to avoid indexing used of expressionsin

an irrelevant context (for example use of "time-of-flight" in relation with travelling), and/or to index separately the use of some terms in different contexts. This classification might also be used to limit the identification of innovation opportunities to relevant area of interests, and improve the speed of the method.

The data analysis component 200 comprises a parser 201, a lifecycle analyser 202, at least one innovation model 203and a semantic new term scanner 204. Those modules are preferably built as software modules carried out by a processing unit. The parser 201 and the semantic new term scanner 204 seek to automatically identifynew expressionsappeared in the data collections 101,102 and/or 103 during the period of interest.A new expression is an expression which appeared from the first time during a given time period, or whose use (trend data) as indicated by the lifecycle analyser 202, weighted by the innovation model 203, follows a given pattern during this period.

This determination of new expressionsfrom the data collection component 100 can rely on two complementary strategies. The first of these is pattern-based extraction, whilst the second is based on statistical co-occurrences. The parser 201 and semantic new term scannercomponentscanusea specially defined ontology (see domain modelling component, 300) and a list of expressions stored in the data collection component, 100. An example of ontology related to some aspects of game consoles is illustrated in Figure 3.

The output of the semantic new term scanner 204 and parser 201 areautomatically generated new expressionsthat have the potential to match the ontology in 300, i.e., which are potentially related to the area of interest. Each newexpressionis provided with a link providing a direct access to a list of sources in 101, 102, 103 in which this new expression appears.

The method of operating the data collection module 100 and data analysis component 200 may include

- Gather data from external sources 101, 102, 103 in a database in data collection 100 component
- Concatenatedata in single textual form
- Split content inexpressions, such as bi-grams for example
- Use content filterwhich may include
  - url
  - email
  - domain names
  - non ascii char and punctuation
  - person names
  - upper case
- Tokeniseand Count
- TermsRanker
  - StatisticalRanker
  - ContextRanker
- Writenewexpressionsto data analysis component database

The lifecycle analyser component 202 is a further module, preferably a software module, executed by the processing unit. Its purpose is to determine a likelihood of innovation at each time for each expression, based on the trend data stored in database of collection component 100 and possibly on other data. The lifecycle analyser component uses an innovation model in order to determine the innovation likelihood associated with an expression, as a function of time during a period.

An example of data output by the module 202 is illustrated on Figure 8 which

shows the innovation likelihood associated with the term "LED backlit LCD" over time. The assumption behind the innovation model used by the lifecycle analyser 202 is that innovation described with a specific expression is much more likely to happen at specific period of time, for example when the use of this expressionstarts to increase in data sources, and then progressively decreases as this expression becomes more common. In one aspect, the innovation model is dependent on the technical domain ("area of interest"). For example, use of new expressions tends to widespread faster in information technology than in medical technology. Therefore, a purpose of the lifecycle analyser is to provide, in an automatic way, data useful for identifying which of the expressions in the database of data collection 100 are "new expressions", i.e., expressions whose use in databases started to increase in a significant way during the period of time selected by the user, whereas the significance of the increase depends on the innovation model for the given technology.

The domain modelling component 300 is a module, such as a software module with a database, which stores reference data describing the area of interest. The description of the area of interest may be expressed as a graph or ontology, which can fully represent an area of interest at each level of its structure. An example of ontology is illustrated on Figure 3. The graph containselements (termsor expressions related to technologies or substances) and relationships between these elements. The tool may also display a visual representation. Optionally, relationships can be explicitly defined (e.g. part_of, is_a) to draft a formal ontology to be represented in XML-like standards, such as OWL.

The reference data may be amended and updated over time by the lifecycle analyser 202 and/or by the user via the user toolbox 500.

The matching engine 400 receives data from the domain modelling component 300 and from the data analysis component 200 and carries out matches between new expressions identified by data analysis component 200 and elements in the reference data and identifies innovation opportunities for presenting to the user via the user toolbox 500. The purpose is to determine, from all the new expressions retrieved by the data analysis module 200, those which might be related to the area of interest as defined by the domain ontology in 300.

The matching is carried out either directly by unifying elements and relationships, or by exploiting some external ontological resources such as those referred to in the domain modelling component 300. In the former strategy, elementsshall be normalized and matching of new expressions with elementsof the ontology might be more or less constrained using string-to-string edit distances or morphological analysis so that morphological variants (e.g. *magnet*, *magnetic*, *magnetism*) can be conflated to the same semantic unit.

One possible method associated with the matching engine 400 may include:

- Load stored reference data from domain modelling component 300 into local store
- Extracting each element
- Select suitable elements
- Loadn-grams found in new expressions as determined by data analysis component 200
- Match each element of the ontology 300 against each tokenised n-gram found in the new expressions

- Calculate possible matches
- Create for each good match a new annotation to the associatedreference data term

The user toolbox 500 presentsresults to an end user in an interactive way on either fixed or mobile platforms.Presentation of the results may be done visually or via an audio output, for example. This is done in such a way so as to allow the results to be logically interpreted and substantiated. For example, a portion of the reference data may be displayed showing an element against which a new expressionhas been matched. Users may click on a representation indicating that there has been a match, and see anyunderlying innovation likelihood measure. Other related information may be provided such as news articles, blogs, etc. Notification of convergence matches can occur in real-time to the end-user through multiple channels, such as text message and e-mail.

Referring now to Figure 2, in some aspects of the invention reference data stored in module 300 and describing elements (such as terms or expressions) relating to an area of interest is provided by a reference framework (or ontology) which gives a broad semantic definition of the area of interest. At step 10 the reference framework is created and stored for use in the innovation generation method of the present invention.

Thestored reference data is represented as an ontologyby the reference framework. This is a system of interconnected nodes each node describing an element of the area of interest. An example is illustrated on Figure 3. Figure 3 shows a portion of an example reference framework for a portion of a Games Console. In the embodiment described here the area of interest is a particular product. Each of the elements which make up the product is defined by a node usually representing a semantic expression (which may be a single word

or a phrase). Anysub-elements which make up an element aredefined by related nodes etc. until a complete structure is defined. Conveniently this structure may be a tree or tree like and may be modelled on forms traditional in ontologies, but it will be appreciated that the interconnections between nodes are not limited to such a hierarchical structure. Optionally, further underlying elements or characteristics for each node may be defined as can be seen in the example shown in Figure 3.

At step 20 expressionsare selected for analysis.Expressionsmay be selected by various criteria. Primarily however this may be by identifying in the data analysis component 200 newexpressions (terms or expressions not previously known and not present in the reference framework).

The analysis of each expressionis carried out at steps 30 and 40 by the Lifecycle Analyser which takes the selected expression $T_i$and outputs a time dependantinnovation likelihood measureof the expressionbased on trend data stored in component 100.

The innovation likelihood measure is in the form of a number or a probability which represents a likelihood that the expression Timight contribute to a novel invention at a given time t_j.

At step 30 the Lifecycle Analyser thus retrievestrend data for the expression$T_i$over time for example, for example based on search volume on an Intranet or the Internet (using a tool such as Google Insight for example), or the number of hits reported by a web crawler (such as a web crawler part of component 100, or any other crawler) or the frequency of the expression in searched documents. At step 40, thetrend datathus determined is then used to infer a time-series innovation likelihoodof the element. The Innovation

Model is then applied to determine an innovation likelihood measure for the expressionT_i, i.e.,the probability of the expression$T_i$contributing to a novel invention.The Innovation Model defines the relationship between the trend data and inferred likelihood of invention novelty, both of which are usually time variant.

Depending on its source, the trend data may be noisy, and data from different sources may have different amplitudes. The trend data thus acquired is pre-processed to clean the data by smoothing and normalisation algorithms.

For example, such time series data may be processed further to make it easier to identify time-series trends. Functions used may include time domain signal processing functions. Alternatively, or additionally,conversion into the frequency domain may be carried out using a Fourier Transform to provide further differentiation between possible low/high value contributors to inventions and the basis further processing using signal processing functions for further analysis.

Such transformations can allowdetermination of innovation probabilities that depend upon the rate of change, or cycles, of the trend data. For example, if a Fourier Transform is used different frequency bands will correspond to different 'frequency components' in the time based trend data, which will depend upon the rate of change of the time based trend data.

Figure 4 illustrates trend data in the form of processed and scaled word frequency data against time for the semantic expressions “time of flight” and “led backlight”.

The trend data is input to an Innovation Model such as the one illustrated in

Figure 5 to produce aninnovation likelihood which varies over time (i.e. the output is related to the probability of the element contributing towards a novelinnovation opportunity).The Innovation Model represents a transfer functionwhich mapstrend data to a likelihood that a novel innovation will result.In a further example illustrated in 4a, the result of applying an Innovation Model to the trend data illustrated in Figure 4 can be seen. As can be seen a policy of the Innovation Model has been defined such that when the trend data for the expression "LED backlight" in Figure 4 exceeds a defined threshold for the first time then it is inferred that inventions related to this expression will have the highest likelihood of novelty. This decreases over time, which is reflected by the function which is then applied. The gradient and indeed the timing of the original threshold point may be linked to the contextual match against the stored reference data. For example, fast changing industries are likely to have a higher gradient than those which change more slowly. Because of this the characteristics of the Innovation Model are indicative of the speed at which information flows in the area of interest and the efficiency with which this information is applied by those who work in that area.

The Innovation Model may be simple - for example, it may apply a logarithmic scaling factor to the trend data, starting from when the trend first exceeds a certain threshold (the start of the lifecycle or 'S' curve). Alternatively integral / differential or other functions may be used according to the relationship between innovation value and the trend data.

Innovation models can be manually derived, or automatically calculated by making reference to the frequency and evolution of innovations within a specific domain.For example a rate of innovation may be determined by determining the time between anexpression first appearing in a data search

and a priority date for a patent covering an associated innovation.

The innovation likelihood value data is used to derive an innovation likelihood measure for each selected expressionT_i at a given time.

At step 50 potential innovation opportunities are identified and presented to the user by the user toolbox 500 on the basis of the innovation likelihood measures determined by the Lifecycle Analyser at step 40. The list of innovation opportunities might be a ranked list, based on various possible ranking criteria, including for example the innovation likelihood.

Some aspects of the invention include a feedback step. At step 60 a usermay indicate which matches are relevant, and at step 70 this information is used to update the reference data(the ontology) so as to adaptthe system over time.This may be done by updating the ontology / reference framework.The parameters of the innovation model may also be updated.

In oneaspect of the invention new expressions identified by the semantic new term scanner 204 may be matched to the reference framework 300 and only those new expressionshaving a match in the reference framework may be processed. It is also possible to process only new terms found in documents classified in relevant classifications.

With reference to Figure 6, at step 501 the semantic new term scanner 204 is used to identify new expressionsfrom data passed from the data collection component 100.

The new term scanner 204 attempts to generate new synonyms (acronyms), as well as hypernyms(superordinate word or phase) and hyponyms (subordinate

word or phrase) of expressions collected from external sources 101-103.

At steps502 and 503, innovation likelihood measuresare generated for new semantic expressionsby the Lifecycle analyser202 and these are stored together with the new semantic expressions.

At step 504the new expressionsand their innovation likelihood measuresare compared with the semantic elements represented by nodes of the reference framework 300 together with the innovation likelihood measuresof those elements, in the matching engine 400 to determine a match. Such matching might, in some aspects, only be done for nodes which have already been identified as being of particular interestaccording to the own innovation likelihood measure. For example, this matching might be done for nodes which have themselves a high innovation likelihood.

Each new expressionmay match against more than one element of the reference data, or it may be the case that the new term scanner 204 produces the same new semantic expressionfor more than one existing expression. This event is recognized by the matchingengine as a multiple match event. When a multiple match event occurs, the new expression is identified as being matched against a plurality of elements in the reference framework 300.

At step 505 further analysis may be carried out to compare the innovation likelihood measuresof matched new expressionswith those of elements both above and below the matched elements in the reference framework hierarchy so as to be able to understand how an innovation may flow across the components which make up the area of interest, and how it will relate directly to the area of interest itself.

Such further analysis may also identify potential innovation opportunities.

In a second embodiment of the invention, at step 20 the new term scanner 204 is used to identify new expressions$T_i$which may be of interest in the area. The new term scanner 204 does not search for new expressionswhich are related to those in the reference frame work, rather the new term scanner searches for new expressions'per se' i.e.new or recently introduced expressions such as new words or bi-grams not present, or not as frequent before in sources 101-103.
Trend data for those new expressions$T_i$is collected from a data source and the trend data is pre-processed and normalised as described above.

In this aspect of the invention selected expressionsmay relate to an unknown area of interest and therefore a default Innovation Model may be used to determine innovation likelihood measures at step 40.

Figure 7 is an illustration of various embodiments of the present invention, which illustrate how the Lifecycle Analyser may either be applied to new expressionsfirst or to expressions of the stored reference data first. In one aspect of the invention 701, expressions are selected, the lifecycle analyser is applied and the results are presented to the user (corresponding to steps 20, 30, 40, 50 on Figure 2).

In the first embodiment of the invention 703 new expressionsare identified at step 703 and the lifecycle analyser 202 is applied to those. In one variation 704 the lifecycle analyser is only applied to new expressions that match the reference framework 300. In a preferred variation of that embodiment 705 the lifecycle analyser is then applied to expressions selected from the stored reference data.

In the second embodiment of the invention 702, expressions are selected from the stored reference data as described previously. In a preferred variation of that embodiment 706 the lifecycle analyser is then applied to new expressionsthat match the stored reference data.

In either aspects, once the lifecycle analyser has been applied to both expressions selected from stored reference data 300 and to new expressions determined by parser 201 and/or semantic new term scanner 204, an embodiment 707 includes the step of identifying multiple matches.

An embodiment of the invention may be exemplified as follows. With reference to Fig. 3, the expression “LED” is selected from the stored reference data. A new expression, “LED backlit LCD” which is semantically related to the stored element “LED” is then identified. Trend data which may, for example, be based on word frequency publications for this new expression is determined as shown in Fig. 4. An innovation likelihood measure is then determined, as shown in Fig. 4a for the expression “LED backlit LCD”. Optionally, this trend data may be determined contextually by considering those elements of the stored reference data of which “LED” forms a part. An innovation likelihood measure is also determined for the expression “LED” and to exclude the use of LED in different contexts. Innovation likelihood measures for both the terms of “LED” and “LED backlit LCD” are then used to determine the likelihood of an innovation opportunity at the time when the new element is identified. Accordingly in this example, there is a match between the LED used in a Time-of-Flight gesture recognition system and a LED backlit LCD. This indicates the possible basis of an invention. The likely novelty of this is then determined by the innovation likelihood measure of both expressions.

Another embodiment of the invention may be exemplified as follows. This example relate to new use of LEDs for producing light used in a time-of-flight based 3D light sensor. With reference to Fig. 3, the expression “LED” is selected from the stored reference data.A new expression, “LED backlit LCD” is then identified by the semantic new term scanner 204. Trend data which may, for example, be based on word frequency in selected publications for this expression is determined as shown in Fig. 4. An innovation likelihood measure is then determined, as shown in Fig. 4a by applying an innovation model which defines a relationship between the time variant trend data and the innovation likelihood measure. The innovation likelihood measure is then used to determine the likelihood of an innovation opportunity at a particular point in time for the trend data. In this example, there is a match between the “LED” used in a Time-of-Flight gesture recognition system and a “LED backlit LCD”. This indicates the possible basis of an invention. The likely novelty of this is then determined by the innovation likelihood measure, and the likely novelty of the match subsequently substantiated.

Trend datawhich may, for example, be based on expressionfrequency in selected publications is calculated for a plurality of new expressions. One of these is the term “LED backlit LCD”. An innovation likelihood measure is then determined, as shown in Fig. 4a, which determines that this expression may result in a novel invention. The expression “LED backlit LCD” is then matched against the stored reference data, which may be an ontology, in this case producing a semantic match against the expression LED to produce the innovation opportunity.

The components described herein may be implemented in a variety of ways. Referring to FIG. 1, various modules 100, 200, 300, 400 and/or 500 are

represented as a series of interrelated functional blocks that may represent functions implemented by, for example, one or more integrated circuits (e.g., an ASIC) or may be implemented in some other manner as taught herein.

A database, such as for example the database which might be included in the data collection 100, the domain modelling 300, etc. might be implemented as a relational database using a SGBD, as collection of documents such as for example XML documents, as a collection of text files, as a semantic database or filestore, as a cloud store, or with any other suitable data storing means.

The apparatus might comprise one single, integrated apparatus, such as a computer or server or set of servers, or a plurality of interconnected equipment, including equipment interconnected over an Internet, over the Internet, or in a cloud.

Eachmodule 100, 200, 300, 400 and/or 500 may include one or more modules that may perform one or more of the functions or steps described above with regard to various figures. For example, the data analysis module 200 may comprise a plurality of sub-modules 201, 202, 203, 204. In addition, eachmodule 100, 200, 300, 400 and/or 500 represented by FIG. 1 as well as other components and functions described herein, may be implemented using any suitable means, including as a software module carried out by suitable processing means, such as a processing unit, for carrying out the required functionality and steps. Each step described and claimed in the present application may be carried out by a suitable software module executed by suitable processing unit for carrying out the required functionality.

The methods described and claimed may be carried, at least in part, using software modules carried out by a suitable processing unit or other processing means.Those of skill would further appreciate that any of the various

illustrative logical blocks, modules, processors, means, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, various forms of program or design code incorporating instructions (which may be referred to herein, for convenience, as "software" or "module" or "software module"), or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality.

A software module (e.g., including executable instructions and related data) and other data or database may reside in a data memory such as RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer-readable storage medium known in the art.
Also, it should be understood that any reference to an element herein using a designation such as "first," "second," and so forth does not generally limit the quantity or order of those elements. Rather, these designations may be used herein as a convenient method of distinguishing between two or more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements may be employed there or that the first element must precede the second element in some manner.
It is understood that any specific order or hierarchy of steps in any disclosed process is an example. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the processes may be rearranged while remaining within the scope of the present disclosure. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented.
Moreover, in some aspects any suitable computer-program product may comprise a computer-readable medium comprising codes (e.g., executable by

at least one computer) relating to one or more of the aspects of the disclosure. A computer program product may comprise packaging materials.

CLAIMS

1. A computer-enabled method of determining innovation opportunities comprising the steps of:

a) causing a data collection module (200) in a processing unit to store expressions retrieved from external data sources (101, 102, 103) in a data collection (101, 102, 102);

b) causing a data analysis module in a processing unit to identify new expressions among said expressions, based on trend data retrieved for the selected expressions, said trend data comprising use of said expressionsin said data source during a period;

c) causing a module (400) in the processing unit to identify (50) an innovation opportunity associated with the identified new expressions; and

d) presenting (50) said innovation opportunity for information and analysis by a user.

2. A method according to claim 1, further comprising:

selecting an area of interest;

matching the identified new expressions with the selected area of interest;

presenting said innovation opportunity only for new expressions which match said selected area of interest.

3. A method according to claim 2, comprising the steps of:

modelling said area of interest with an ontology (300);

semantically matching the identified new expressions with elements of said ontology,

presenting said innovation opportunity only for new expressions

which semantically match said elements of said ontology.

4. A method according to one of the claims 1 to 3, in which the step of selecting one or more new expressions comprises providing a link to one or more documents or pages used in the determination of said new expressions.

5. A method according to claim 4, wherein a patent classification is used to automatically exclude expressions which are found in documents not related to the area of interest.

6. A method according toone of the claims 1 to 6, further comprising a step of causing a lifecycle analyser (202) in said processing unit to apply a transfer function to said trend data and determine an innovation likelihood measure for a stored expression,

wherein an expression is identified as anew expression depending on said innovation likelihood measure for said expression.

7. A method according to claim 6, wherein said innovation likelihood computed by said lifecycle analyser (202) is a function of time.

8. A method according to claim 7, wherein the application of said transfer function results in associating to said expression the highest innovation likelihood at the instant when said trend data for an expression exceeds a defined threshold for the first time, and associating to said expression a decreasing probability after said instant.

9. A method according to claim 7, wherein said transfer function is domain specific.

10. A method according to one of the claims 1 to 9, further comprising a step of causing a lifecycle analyser (202) in said processing unit to apply a transfer function to said trend data and determine an innovation likelihood measure for a new expression,

wherein said innovation opportunity is determined based on said innovation likelihood measure for said new expression.

11. A method according to claim 10, wherein said innovation likelihood computed by said lifecycle analyser (202) is a function of time.

12. A method according to claim 11, wherein the application of said transfer function results in associating to said expression the highest innovation likelihood at the instant when said trend data for an expression exceeds a defined threshold for the first time, and associating to said expression a decreasing probability after said instant.

13. A method according to claim 11, wherein said transfer function is domain specific.

14. A method according to one of the claims 3 to 13, further comprising a step of causing a lifecycle analyser (202) in said processing unit to apply a transfer function to trend data associated with elements of said ontology (300), and determine an innovation likelihood measure for said elements,

wherein said innovation opportunity is determined based on said innovation likelihood measure for said element.

15. A method according to one of the claims 3 to 14, further comprising the step of:

updating the ontology (300) in dependence upon feedback from a user relating to said innovation opportunity.

16. A method according to one of the claims 1 to 16, further comprising the step of:

causing a module in the processing unit to rank a plurality of innovation opportunities based on an innovation likelihood at a given instant.

17. A method according to one of the claims 1 to 16, wherein said trend data comprises a number of searches for saidexpressionwith a search engine over a time period.

18. A method according to one of the claims 1 to 17, further comprising presenting to the user the innovation opportunitiesasa portion of semantic reference data along witha new expression.

19. A method according to one of the claims 1 to 18, further comprising presenting to the user each innovation opportunity along witha link to source documents where new expressions related to this innovation opportunity have been found.

20. An apparatus for determining innovation opportunities comprisingcomputingmeans for selecting one or more expressions, and alifecycle analyser (202) arranged in operation to:

-determine trend data relating to a selected expression;

-determine an innovation likelihood measure for said selected expression in dependence upon said trend data;

-identify an innovation opportunity in dependence upon said innovation likelihood measure ; and

-presentation means for presentingsaid innovation opportunity for information and analysis by a user.

21. An apparatus according to claim 20, in which the means for selecting one or more expressions comprises a new term scanner arranged in operation to identify one or more new expressions.

22. An apparatus according to one of the claims 20 or 21, further comprisinga new term scanner (204) arranged in operation to identify one or more new expressions related to stored reference data;and in which the lifecycle analyser (202)is further arranged in operation to

-determine trend data relating to at least one of said new expressions;
-determine a new expression innovation likelihood measure for said at least one new expressionin dependence upon said trend data;
-and identify the innovation opportunity in further dependence upon the new expression innovation likelihood measure.

23. An apparatus according to claim 22, further comprising:

means for identifying a multiple match between a new expression and an expressionin said stored reference data.

24. An apparatus according to claim 22, in which said reference data comprises an ontology (300) comprising a plurality of interconnected nodes, each node describing an element and thus representing a characteristic of an area of interest.

25. An apparatus according to claim 22, further comprising means for updating the stored reference data in dependence upon feedback from a user relating to said innovation opportunity.

26. An apparatus according to claim 21, in which the lifecycle analyser (202) is arranged to use a domain-specific innovation model arranged in operation to generate said innovation likelihood measure.

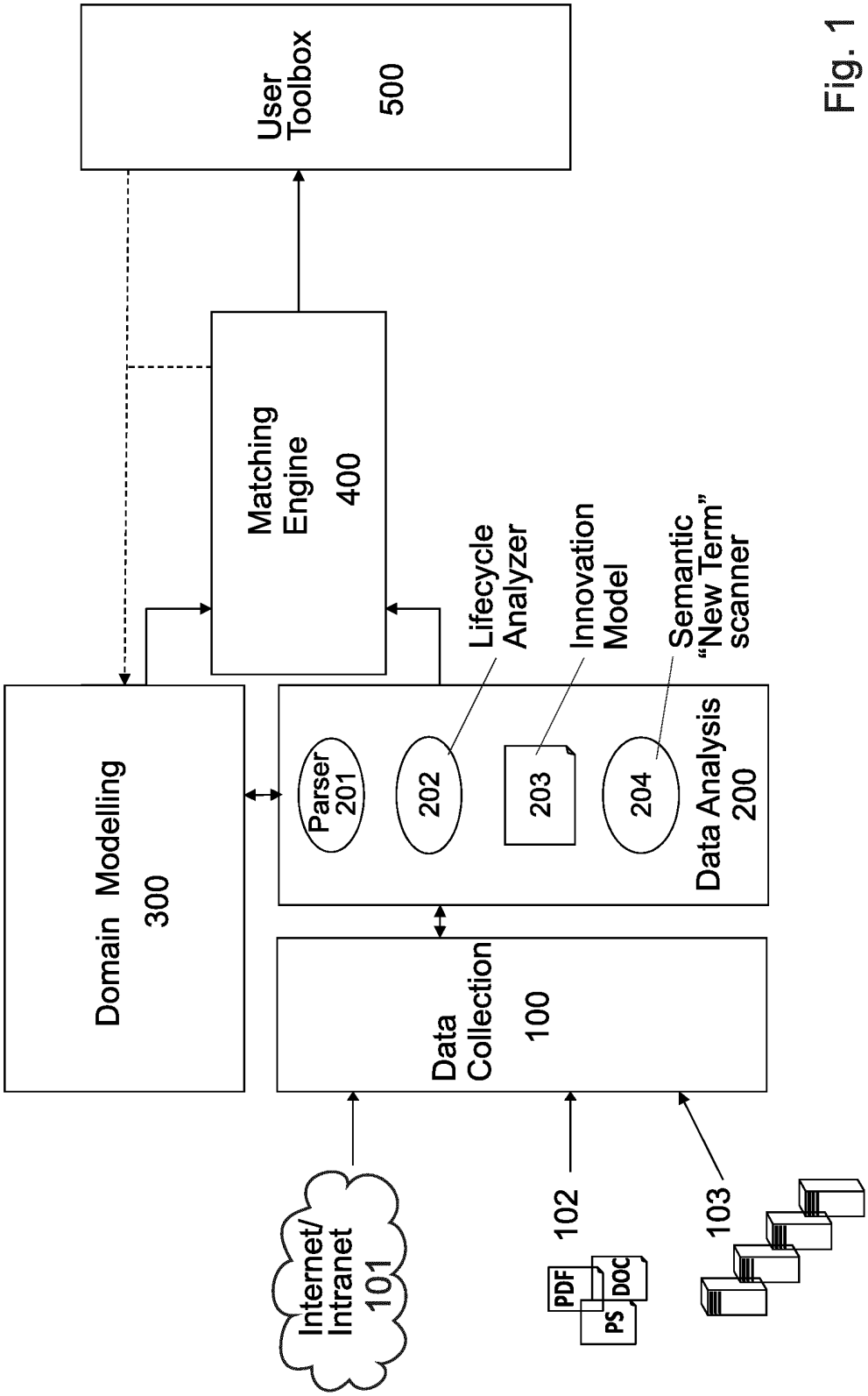
User Toolbox
500
Matching Engine
400
Domain Modelling
300
Parser
201
202
203
204
Lifecycle Analyzer
Innovation Model
Semantic "New Term" scanner
Data Analysis
200
Data Collection
100
Internet/ Intranet
101
102
PDF
DOC
PS
103

Fig. 1

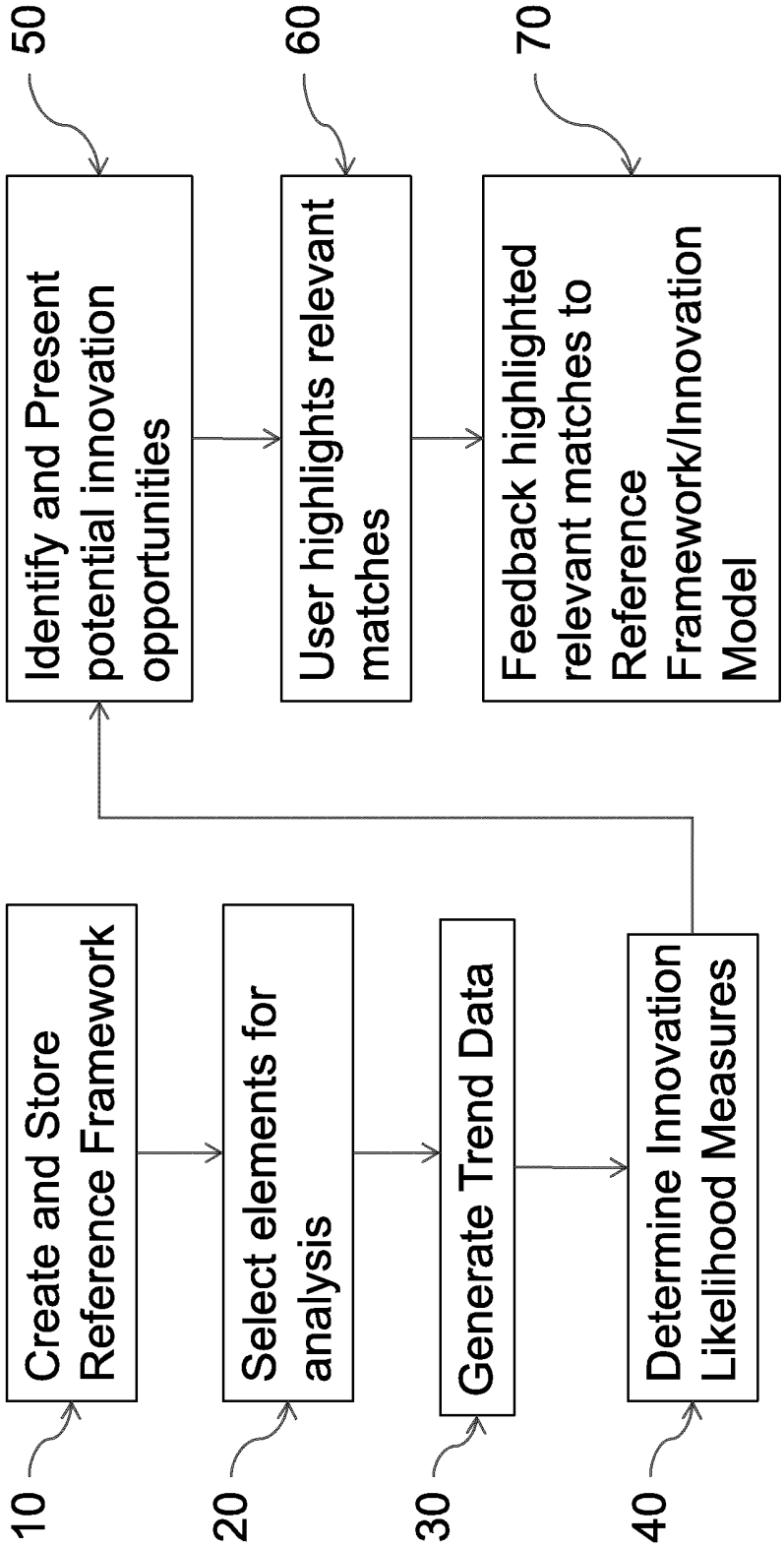
10
Create and Store Reference Framework
20
Select elements for analysis
30
Generate Trend Data
40
Determine Innovation Likelihood Measures
50
Identify and Present potential innovation opportunities
60
User highlights relevant matches
70
Feedback highlighted relevant matches to Reference Framework/Innovation Model

Fig. 2

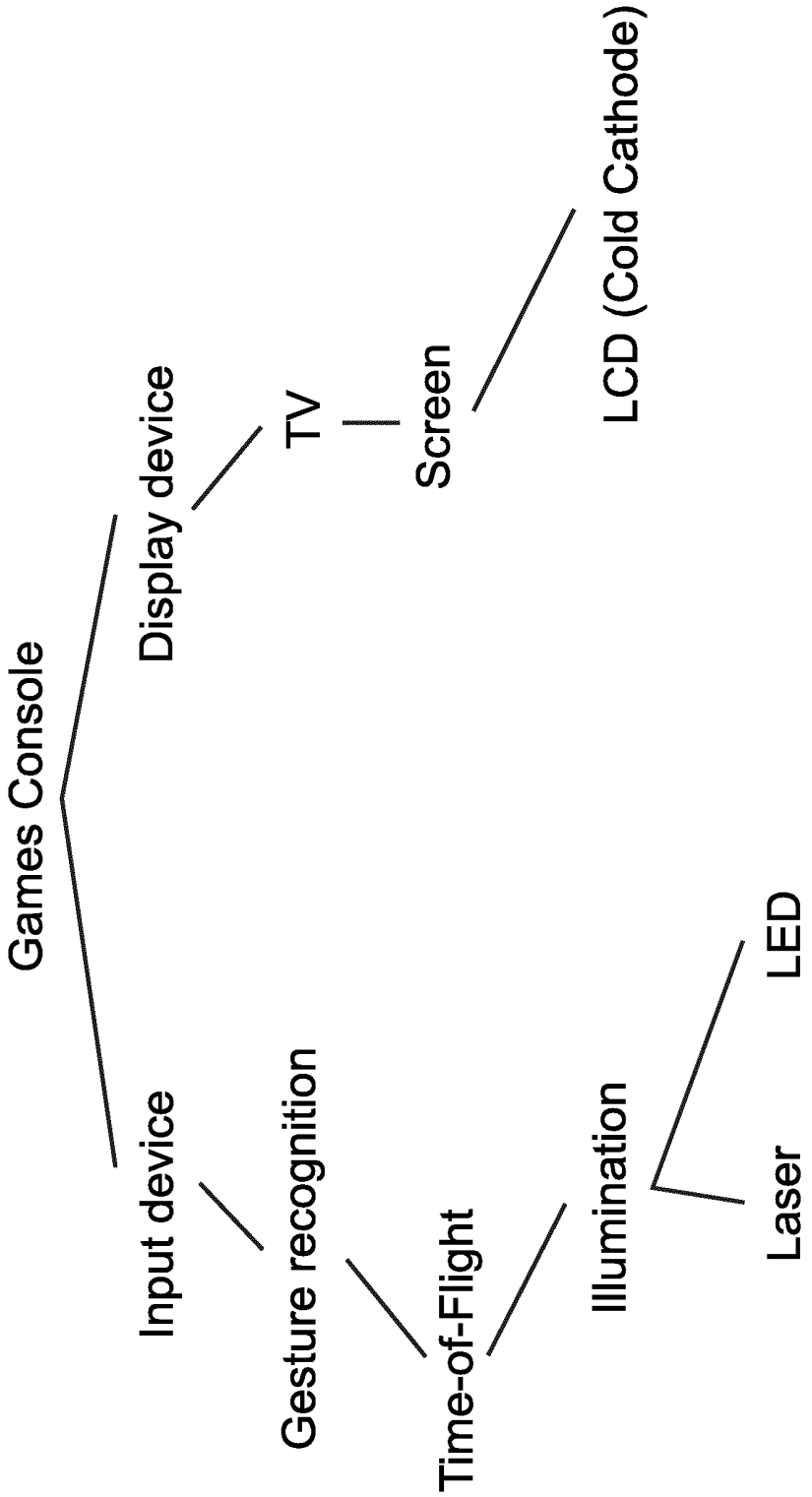
Games Console
Input device
Display device
Gesture recognition
TV
Time-of-Flight
Screen
Illumination
LCD (Cold Cathode)
Laser
LED

Fig. 3

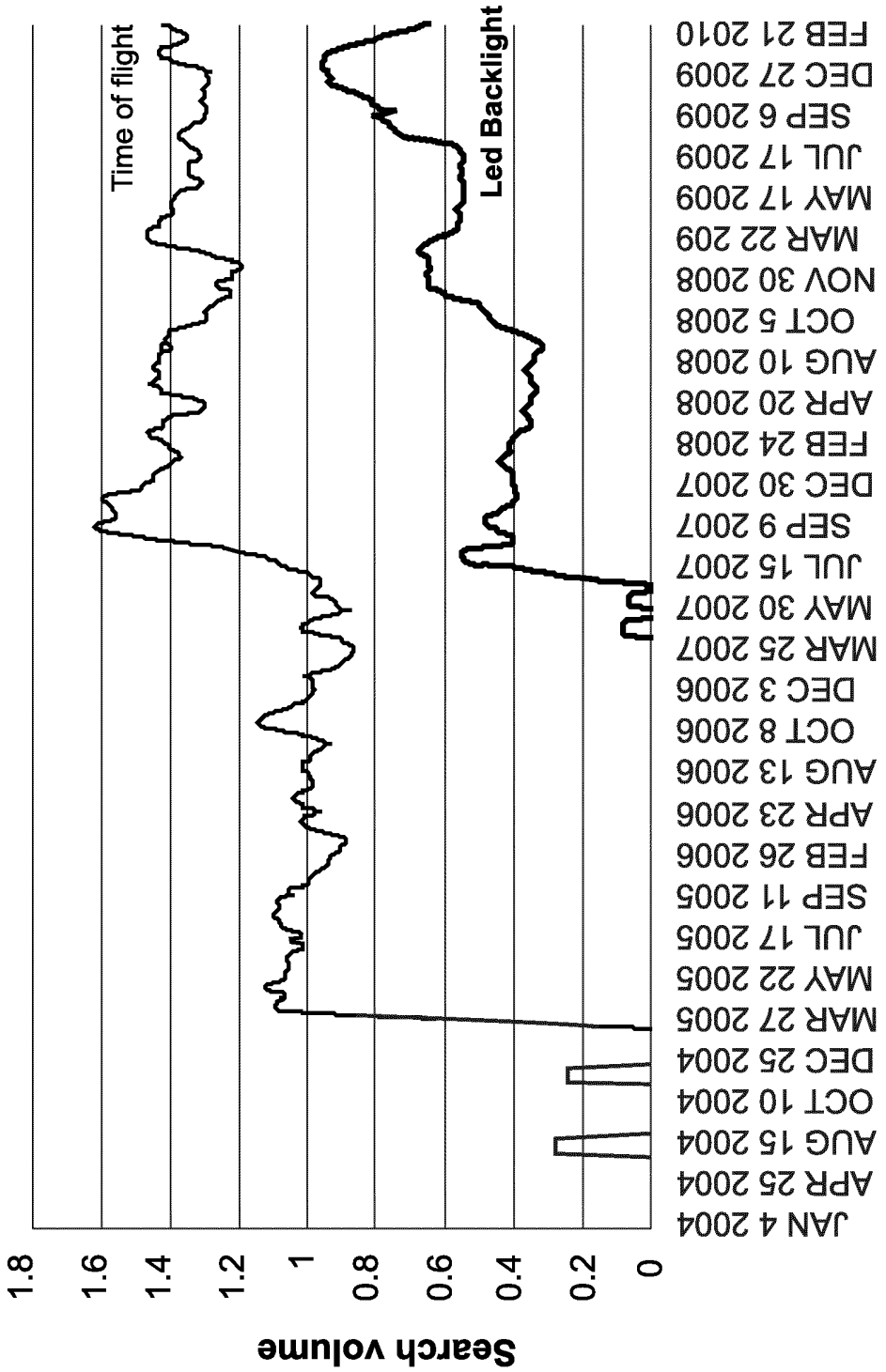
Search volume
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Time of flight
Led Backlight
JAN 4 2004
APR 25 2004
AUG 15 2004
OCT 10 2004
DEC 25 2004
MAR 27 2005
MAY 22 2005
JUL 17 2005
SEP 11 2005
FEB 26 2006
APR 23 2006
AUG 13 2006
OCT 8 2006
DEC 3 2006
MAR 25 2007
MAY 30 2007
JUL 15 2007
SEP 9 2007
DEC 30 2007
FEB 24 2008
APR 20 2008
AUG 10 2008
OCT 5 2008
NOV 30 2008
MAR 22 209
MAY 17 2009
JUL 17 2009
SEP 6 2009
DEC 27 2009
FEB 21 2010

Fig. 4

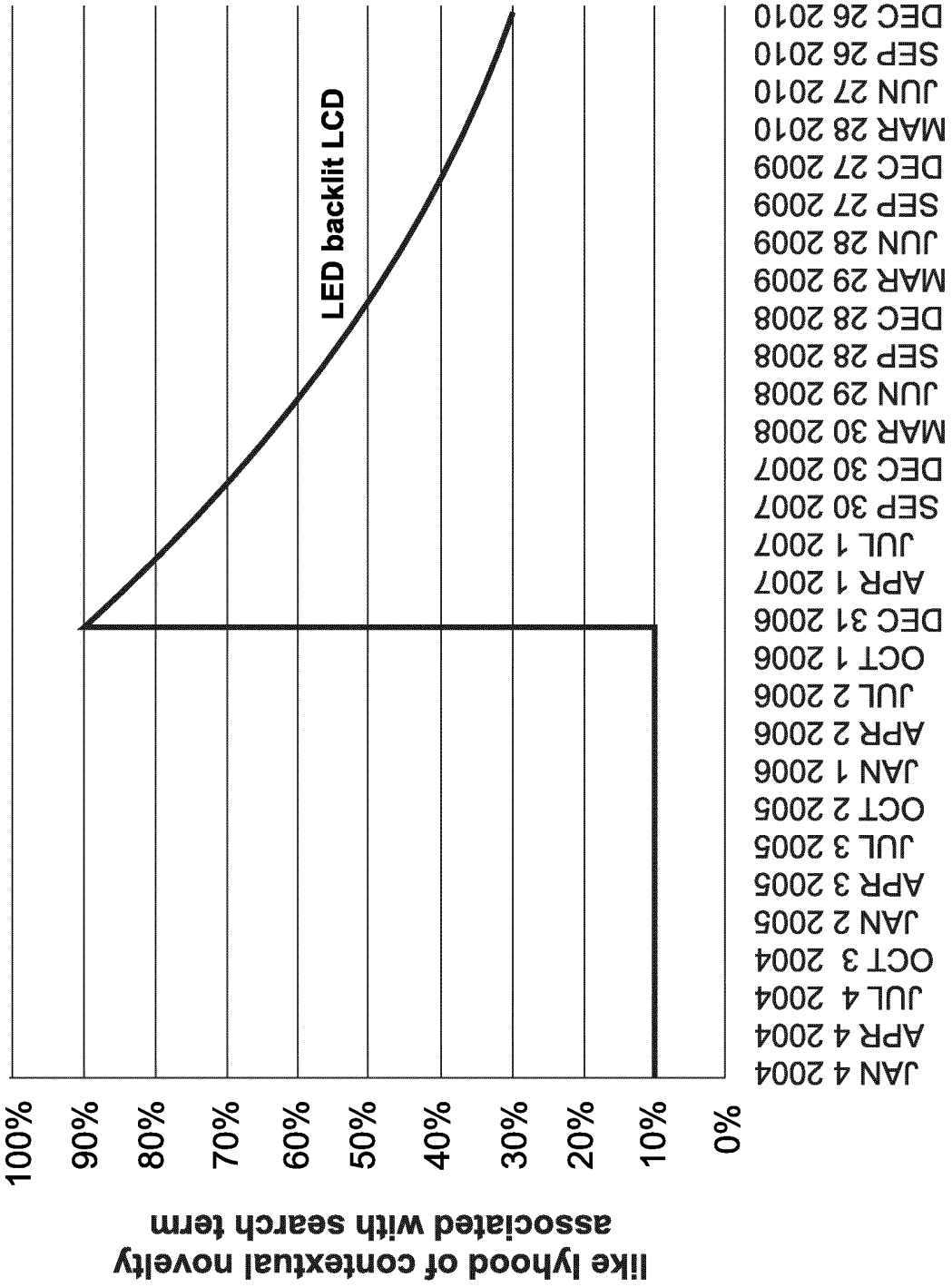
Like lyhood of contextual novelty associated with search term
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
LED backlit LCD
JAN 4 2004
APR 4 2004
JUL 4 2004
OCT 3 2004
JAN 2 2005
APR 3 2005
JUL 3 2005
OCT 2 2005
JAN 1 2006
APR 2 2006
JUL 2 2006
OCT 1 2006
DEC 31 2006
APR 1 2007
JUL 1 2007
SEP 30 2007
DEC 30 2007
MAR 30 2008
JUN 29 2008
SEP 28 2008
DEC 28 2008
MAR 29 2009
JUN 28 2009
SEP 27 2009
DEC 27 2009
MAR 28 2010
JUN 27 2010
SEP 26 2010
DEC 26 2010

Fig. 4a

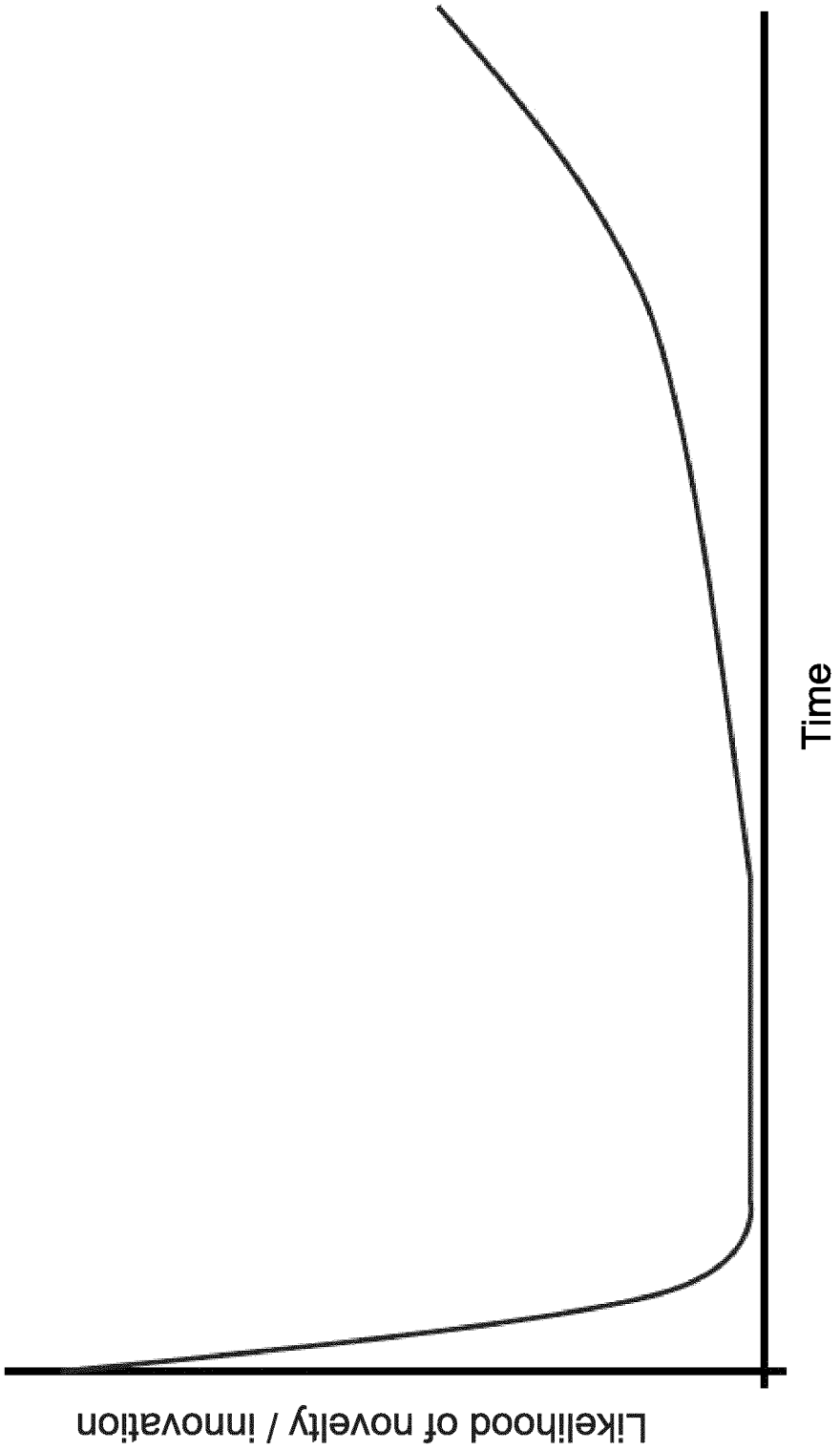

Fig. 5

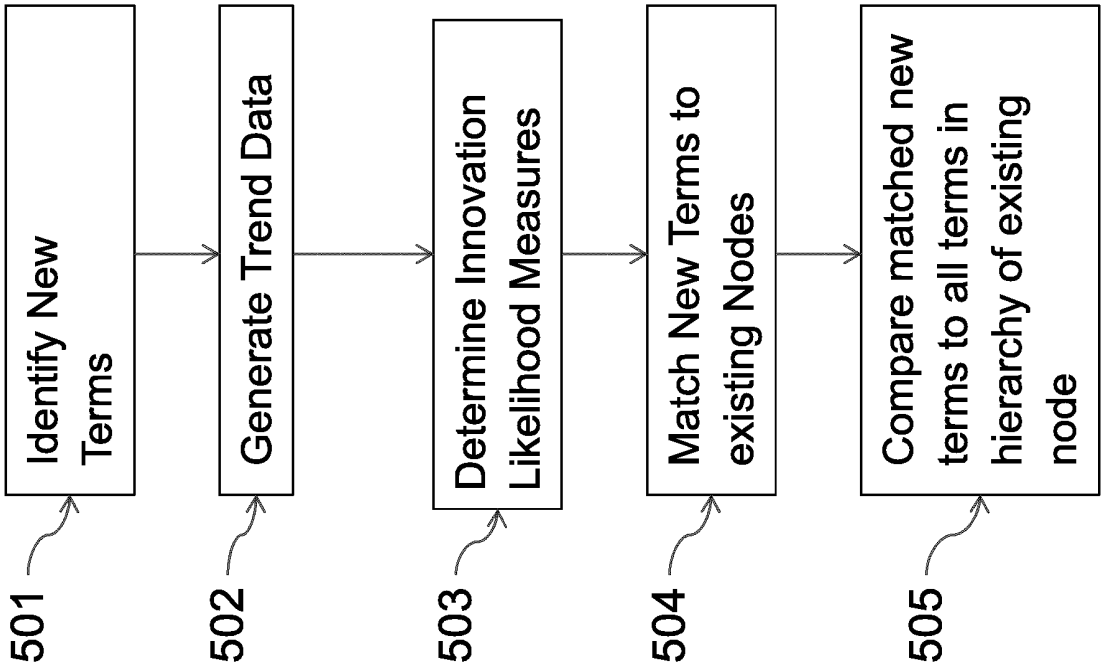
501
Identify New Terms
502
Generate Trend Data
503
Determine Innovation Likelihood Measures
504
Match New Terms to existing Nodes
505
Compare matched new terms to all terms in hierarchy of existing node

Fig. 6

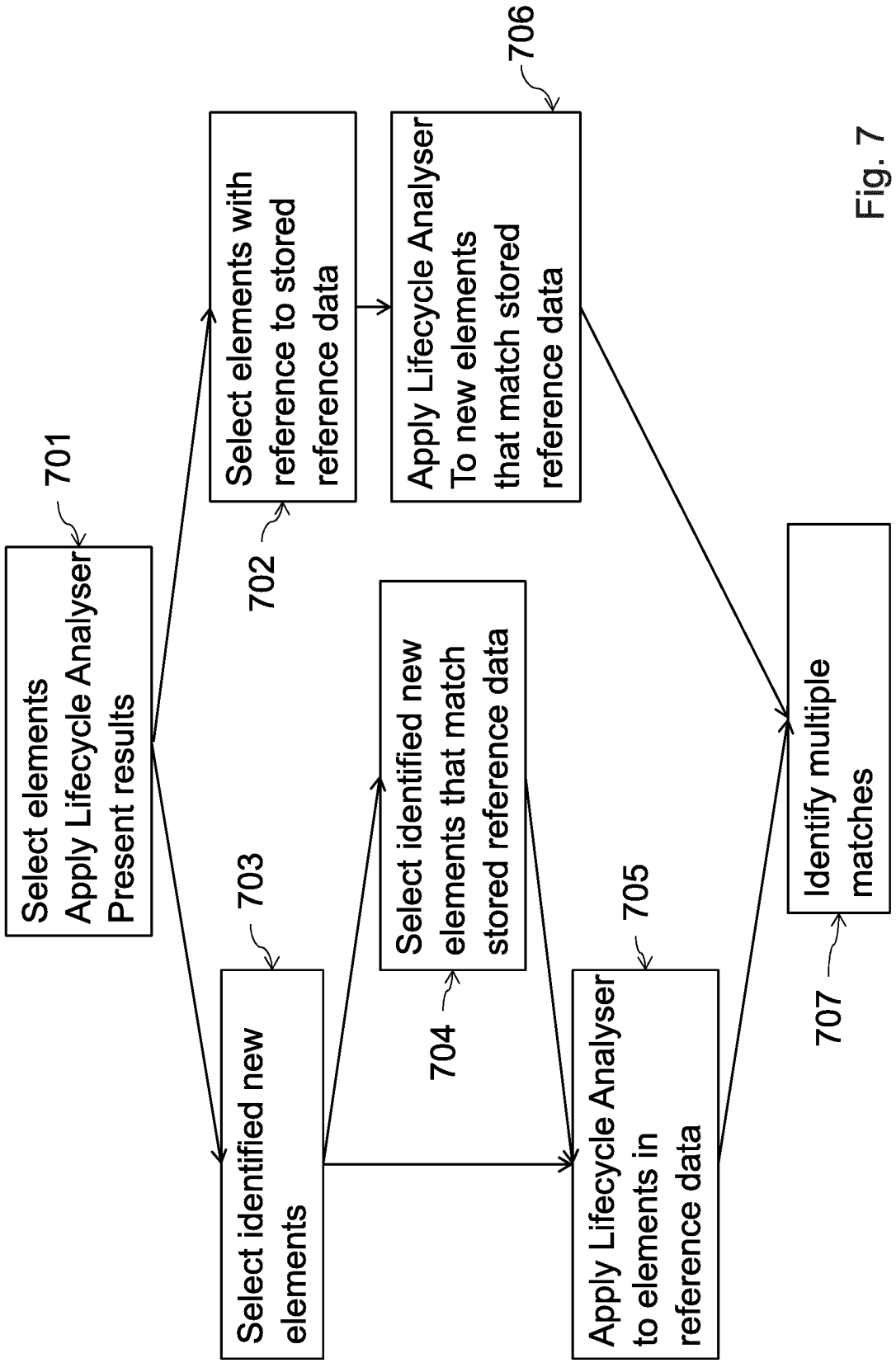
701
Select elements
Apply Lifecycle Analyser
Present results
702
Select elements with reference to stored reference data
703
Select identified new elements
704
Select identified new elements that match stored reference data
705
Apply Lifecycle Analyser to elements in reference data
706
Apply Lifecycle Analyser To new elements that match stored reference data
707
Identify multiple matches

Fig. 7

Fig. 8

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2013/056551

A. CLASSIFICATION OF SUBJECT MATTER
INV. G06F17/30 G06Q10/06 G06Q10/10 G06Q30/02
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G06Q G06F

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	ALAN L. PORTER ET AL: "Technology opportunities analysis", TECHNOLOGICAL FORECASTING AND SOCIAL CHANGE, vol. 49, no. 3, 1 July 1995 (1995-07-01), pages 237-255, XP055066506, ISSN: 0040-1625, DOI: 10.1016/0040-1625(95)00022-3 the whole document	1-26

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[X] Further documents are listed in the continuation of Box C. [ ] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
14 June 2013	24/06/2013

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Dumitrescu, Cristina

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Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2013/056551

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WANG JIWU ET AL: "Empirical research on the technology opportunities analysis based on morphology analysis and conjoint analysis", PORTLAND INTERNATIONAL CONFERENCE ON MANAGEMENT OF ENGINEERING & TECHNOLOGY, 2008 : PICMET 2008 ; CAPE TOWN, SOUTH AFRICA, 27 - 31 JULY 2008, IEEE, PISCATAWAY, NJ, USA, 27 July 2008 (2008-07-27), pages 2041-2047, XP031306825, ISBN: 978-1-890843-17-5 the whole document -----	1-26
X	HIDENAO ABE ET AL: "Trend detection from large text data", SYSTEMS MAN AND CYBERNETICS (SMC), 2010 IEEE INTERNATIONAL CONFERENCE ON, IEEE, PISCATAWAY, NJ, USA, 10 October 2010 (2010-10-10), pages 310-315, XP031805632, ISBN: 978-1-4244-6586-6 the whole document -----	1-26

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